

Assignment: Medical Insurance Cost Prediction using Linear Regression

Course: Machine Learning (4th Year B.Tech)

Topic: Regression using Linear Regression

Difficulty: Intermediate to Advanced

Objective

In this assignment, students will build a **Linear Regression** model to predict an individual's **medical insurance charges** based on demographic and health-related attributes such as age, BMI, smoking status, number of children, gender, and residential region.

Students will perform the complete Machine Learning pipeline including:

- Data Understanding
 - Exploratory Data Analysis (EDA)
 - Data Cleaning
 - Feature Engineering
 - Feature Encoding
 - Feature Scaling
 - Model Training
 - Model Evaluation
 - Hyperparameter Tuning
 - Business Insights
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Dataset

Link: <https://www.kaggle.com/datasets/mirichoi0218/insurance>

Use the **Medical Cost Personal Dataset**.

Target Variable

charges

- Continuous numerical value representing the individual's medical insurance cost.

Example Features

- age
- sex
- bmi
- children

- smoker
 - region
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Business Problem

Health insurance companies need to estimate the expected medical expenses of customers before issuing insurance policies. Predicting insurance charges accurately helps companies determine appropriate premium amounts, manage financial risks, and design personalized insurance plans.

The objective is to develop a **Linear Regression model** that predicts medical insurance charges using customer demographic and lifestyle information.

Business Insights

Assume you are working as a **Data Scientist in a Health Insurance Company**.

Answer the following questions:

1. Which factors have the greatest impact on medical insurance charges?
 2. Does smoking significantly increase insurance costs?
 3. How does BMI affect insurance expenses?
 4. Does age influence insurance charges?
 5. Which feature contributes the most to predicting insurance charges?
 6. How can insurance companies use these predictions for premium pricing and risk assessment?
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Assignment Tasks

Part 1: Understanding the Dataset

Perform the following:

1. Load the dataset.
2. Display the first 10 rows.
3. Check dataset dimensions.
4. Display column information.
5. Check missing values.
6. Identify numerical and categorical features.
7. Display descriptive statistics.

Deliverables

- Dataset summary
 - Minimum five observations
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Part 2: Exploratory Data Analysis (EDA)

Perform:

- Distribution of target variable (charges)
- Correlation heatmap
- Histograms
- Boxplots
- Scatter plots
- Pairplots (optional)
- Outlier analysis

Write observations:

- Which feature has the strongest correlation with insurance charges?
 - Are there any outliers?
 - Which variables appear most useful for prediction?
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Part 3: Data Preprocessing

Perform:

- Missing value handling (if required)
- Label Encoding / One-Hot Encoding for categorical variables
- Feature Scaling (if applicable)
- Train-Test Split (80:20)

Explain why each preprocessing step is necessary.

Part 4: Linear Regression Model

Train a **Linear Regression** model using **Scikit-Learn**.

Display:

- Model coefficients
- Intercept
- Feature importance based on coefficients

Interpret:

- Positive coefficients
 - Negative coefficients
 - Relationship between independent variables and insurance charges
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Part 5: Model Evaluation (20 Marks)

Evaluate the model using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score (Coefficient of Determination)
- Adjusted R^2 Score
- Predicted vs Actual Plot
- Residual Plot

Also explain:

- Which evaluation metric is most suitable for this regression problem?
 - Why?
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Part 6: Model Improvement / Hyperparameter Tuning

Perform any of the following:

- Feature Selection
- Polynomial Features (Degree 2 or 3)
- Ridge Regression
- Lasso Regression

Compare the results with the baseline Linear Regression model.

Present the comparison in a table.

Part 7: Business Insights

Based on your analysis, answer:

1. Which customer segment is likely to incur the highest medical expenses?
2. How does smoking affect insurance charges?

3. Should insurance companies treat smokers differently while pricing policies?
 4. Which features are the strongest predictors of insurance cost?
 5. Suggest strategies to reduce healthcare costs based on the analysis.
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Report Format

The report should contain:

1. Introduction
2. Problem Statement
3. Dataset Description
4. Exploratory Data Analysis
5. Data Preprocessing
6. Feature Engineering
7. Model Building
8. Model Evaluation
9. Model Improvement
10. Business Insights
11. Conclusion

Maximum Length: 8–10 Pages

Submission Guidelines

Submit the following:

- Jupyter Notebook (.ipynb)
- PDF Report
- README.md
- Dataset (if permitted)
- Images used in the report